Pixel charge per electronics frame

October 28, 2026

Requested 8,000 catalogue events per species from the input samples; processed 1,657 double-beta and 1,552 electron events, limited by available input events. All events and all MC particles reach collection. MC generation step: 0.1 mm (simulation input). Random drift uniform in 10–150 cm; thermal isotropic Gaussian diffusion, 200 V/cm bulk field and 293.15 K; W = 21.9 eV. The diffusion histogram has 100 bins per axis and 3σ padding. Out-of-histogram ions are recorded. Ideal hole/pixel assignment maps each diffusion-bin centre to its containing 2 mm pixel. Frame width is 20 ms (2 mm slices at 0.1 m/s). Time is relative to the first arrival edge, increasing source z is later. Per-hit ion counts are rounded integers. Each event has an independent reproducible drift, diffusion and transverse grid phase. A pulse is a pixel–frame cell with positive ion count. Inclusive, event mean and event maximum include every pulse, with no charge threshold. Every event with pulses enters the minimum histogram, using its true minimum over all positive cells. All four distributions have no charge cut. The event mean is its total pulse-ion sum divided by its number of pulses. Events without pulses have no distribution entries. The statistics table uses the same event populations as each histogram. An empty minimum distribution has unavailable mean/median. All histogram count and ion axes are logarithmic; bins are shared between species. Track-reconstruction decisions do not gate this study. No time-integrated pixel charge is used in these plots. Ion retention is all collected pixel ions divided by produced event ions, before histogram truncation. The aggregate ratio uses sums over events; event mean and median give every event equal weight. Loss is one minus retention, caused here only by diffusion-histogram truncation. The minimum histogram selection introduces no collection or threshold loss. Produced ions use the saved per-hit rounded counts. Diffused ion clouds are saved independently of the collection grid when a cache is enabled. Pixel pitch and frame duration can change without repeating Gaussian draws; changing transport or diffusion-histogram settings selects a different cache. Run times include first-event compilation; initial Julia process startup and final histogram/presentation generation are separate.

Charge accounting

Species	Events	Events with charge	Pixel–frame cells with charge	Collected ions	Outside diffusion histogram
Double beta	1657	1657	2,735,921	185,930,020	27,469
Single electrons	1552	1552	2,821,016	173,632,571	21,758

Ion retention

Species	Pixels included	Aggregate retention [%]	Mean event retention [%]	Median event retention [%]	Aggregate loss [%]
Double beta	All pixels	99.9852	99.9852	99.9857	0.0148
Single electrons	All pixels	99.9875	99.9875	99.9883	0.0125

Ion loss accounting

Species	Outside histogram [ions]	Histogram loss [%]
Double beta	27,469	0.0148
Single electrons	21,758	0.0125

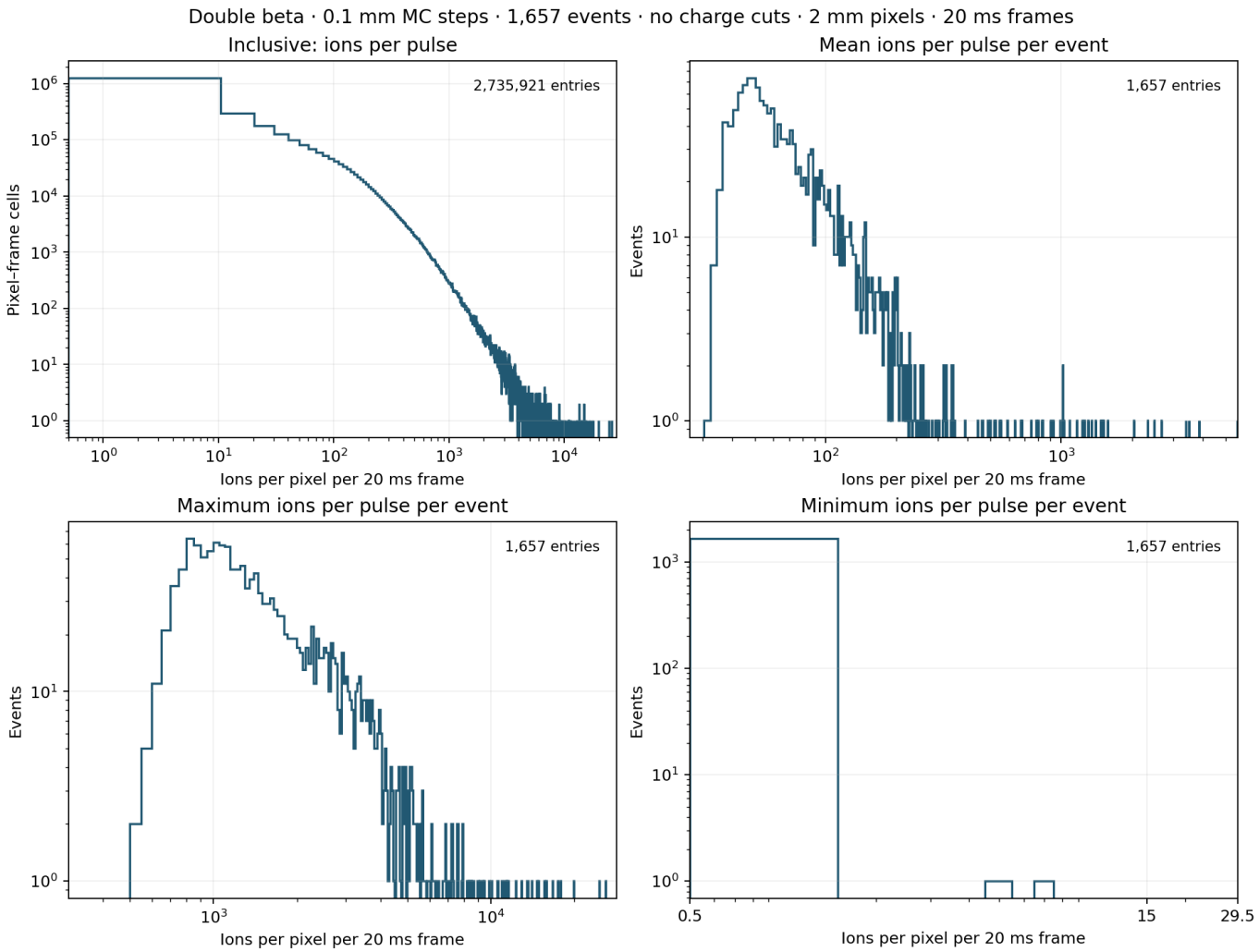
Distribution means and medians

Species	Distribution	Events	Mean [ions/pulse]	Median [ions/pulse]
Double beta	Maximum	1657	2192.87	1534.00
Double beta	Minimum	1657	1.01	1.00
Double beta	Event mean	1657	119.05	67.68
Single electrons	Maximum	1552	2326.86	1594.00
Single electrons	Minimum	1552	1.02	1.00
Single electrons	Event mean	1552	123.86	62.81

Run time

Species	Source read [s]	Diffusion [s]	Cache read [s]	Cache write [s]	Collection/reduction [s]	Total [s]
Double beta	3.199	12.569	0.000	0.000	15.154	32.369
Single electrons	2.720	11.012	0.000	0.000	10.965	25.131

Double beta


Double beta · 0.1 mm MC steps · 1,657 events · no charge cuts · 2 mm pixels · 20 ms frames
Inclusive: ions per pulse
2,735,921 entries
Pixel-frame cells
Ions per pixel per 20 ms frame
Mean ions per pulse per event
1,657 entries
Events
Ions per pixel per 20 ms frame
Maximum ions per pulse per event
1,657 entries
Events
Ions per pixel per 20 ms frame
Minimum ions per pulse per event
1,657 entries
Events
Ions per pixel per 20 ms frame

Single electrons

